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Studierichting: Informatica
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Gegeven is de macht $z = (-\sqrt{3} - i)^{20}$.

Basisgetal omzetten naar poolvorm:

Stel $w = -\sqrt{3} - i$.

$$r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2$$

$$\tan \alpha = \frac{-1}{-\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$\alpha = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$$

$$w = 2 \operatorname{cis} \frac{7\pi}{6}$$

Macht verheffen met De Moivre:

$$z = w^{20} = \left(2 \operatorname{cis} \frac{7\pi}{6}\right)^{20}$$

$$z = 2^{20} \operatorname{cis} \left(20 \cdot \frac{7\pi}{6}\right)$$

$$z = 2^{20} \operatorname{cis} \left(\frac{140\pi}{6}\right) = 2^{20} \operatorname{cis} \left(\frac{70\pi}{3}\right)$$

Argument vereenvoudigen en uitwerken:

$$\frac{70\pi}{3} - \frac{66\pi}{3} = \frac{4\pi}{3}$$

$$z = 2^{20} \operatorname{cis} \frac{4\pi}{3}$$

$$z = 2^{20} \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$$

$$z = 2^{20} \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)$$

$$z = -2^{19} - 2^{19} i\sqrt{3}$$

Antwoord: $z = -2^{19} - 2^{19} i\sqrt{3}$

References